

Digital Archives and Methods Portfolio

Portfolio Exercises and Final Project

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Portfolio overview

This portfolio contains four exercises completed during the course before the final project. The exercises show a progression from data cleaning and regular expressions, through basic R functions and CSV import, to text mining in RStudio. The final project develops these skills further by applying word-frequency and sentiment analysis to protest-song lyrics.

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Submission structure

The portfolio is organised as a continuous submission containing four course exercises, the final project report, and the required metadata. The complete R project, data, figures, and script are available in the accompanying GitHub repository.

The permanent GitHub repository is listed in the software metadata section.

Portfolio Element 1 - Regular Expressions and OpenRefine

This exercise introduced regular expressions and data cleaning in OpenRefine.

1. `\b\s?(\\d{1,2}).\\s?(\\d{1,2}).\\s?(\\d{1,4})` and then substitute using `$3-$2-$1` to get the wanted format of YYYY-MM-DD
2. Firstly, we will wrap each word in quotation marks using the code `^(.+) $`, and then using substitution change the value to `"$1"`. Now each word is in quotations. Next using the code `\\r?\\n` to replace each line break with: `,`. Now each word is in a line with quotations and commas. To turn it back into Voyant style we'll simply find all `"` and substitute them with `blank`. Then we'll find all commas, using the code: `,\\s*` and substitute these with line breaks using `\\n`
3. No, you're simply changing how the data is displayed by using codes. You must make an active choice to remove or alter the data
4. By cleaning and sorting the data in OpenRefine, the answer is October as the driest month with 74 reported as months with no water, and September as the second driest with 70 reported instances
5. I have chosen to look at unmarried women in the age group 20-30. There are in total 2.365 women in this group, and I have excluded the following
 - Sick
 - Not making a living
 - N/A

The list of 10 most frequent jobs in the dataset

- Væver - 37 entries
- Tjenestepige - 28 entries
- Husholder - 19 entries
- Inderste - 13 entries
- Spinder - 13 entries
- Tjener forældrene - 10 entries
- I tjeneste - 9 entries
- Skræder - 7 entries
- Syerske - 7 entries
- Køkkenpige - 6 entries

Portfolio Element 2 - Basic R Functions

Week 10 exercise. This task worked with a numeric vector containing missing values and used basic R functions to inspect and summarise it.

```
rooms <- c(1, 5, 2, 3, 1, NA, 3, 1, 3, 2, NA, 1, 8, 3, 1, 4,
          NA, 1, 3, 1, 2, 1, 7, 1, NA, 4, 3, 1, 7, 2, 1, NA,
          1, 1, 3)
sum(rooms>2, na.rm=TRUE)
#According to the code there are 13 elements exceding 2 rooms

typeof(rooms)
#Double

median(rooms, na.rm = TRUE)
#By using na.rm we can exclude the null results,
#which would beforehand be the median.
#The correct answer is 2
```

The exercise uses `sum()` with a logical condition to count values above two, `typeof()` to inspect the vector type, and `median()` with `na.rm = TRUE` to exclude missing values. The result shows thirteen entries above two and a median of two.

The exercise also clarified the difference between NA and NULL in R. The function `median()` uses `na.rm = TRUE` so that missing values do not affect the calculation.

Portfolio Element 3 - CSV Import and Dataset Inspection

This exercise focused on importing a CSV file and inspecting its structure in RStudio.

Danish Kings

1. Assignment

The separator is commas or ,

2. Assignment

Answer 1:

`read_csv()` is the tidyverse version

Answer 2:

`"spec_tbl_df"` `"tbl_df"` `"tbl"` `"data.frame"`

Answer 3:

The dataset has 11 total columns. This is found by using the function `ncol(kings)`

Answer 4:

#	Name	House	Start_year	End_year	Birth_year	Death_year	Gender	Dynasty	Source	BirthDMY	DeathDMY
1	Gorm den Gamle	Gorm	956	958	908	958	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Gorm_den_Gamle	908	958
2	Harald 1. Blåtand	Gorm	NULL	NULL	936	985	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Harald_Blfatband	936	987
3	Tolke Gormsen	Gorm	985	986	NULL	986	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Tolke_Gormsen	N/A	N/A
4	Svend 1. Tveskæg	Gorm	NULL	NULL	1014	1014	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Svend_Tveskæg	17/04/963	09/02/1014
5	Harald 2.	Gorm	1014	1018	963	NULL	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Harald_2	994	1018
6	Knut 1. den Store	Gorm	1018	1035	NULL	1035	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Knut_den_Store	995	12/11/1035
7	Hårdeknud	Gorm	1035	1042	995	1042	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Hårdeknud	1018	08/06/1042
8	Magnus den Gode	Fairhair	1042	1047	1018	1047	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Magnus_den_Gode	1024	25/10/1047
9	Svend 2. Estridsen	Behniden	1047	1074	1024	1076	M	N/A	https://da.wikipedia.org/wiki/Svend_Estridsen	1019	28/04/1076
10	Harald 3. Men	Behniden	1074	1080	1019	1080	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Harald_Men	1041	17/04/1080
11	Knut 2. den Hellige	Behniden	1080	1086	1041	1086	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Knut_den_Hellige	1043	10/07/1086
12	Oluf 1. Høngar	Behniden	1086	1095	1043	1095	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Oluf_Høngar	1050	18/08/1095
13	Erik 1. Ejegod	Behniden	1095	1105	1050	1105	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Erik_Ejegod	1055	10/07/1105
14	Nels	Behniden	1104	1134	1055	1134	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Nels_of_Denmark	1134	25/06/1134
15	Erik 2. Emune	Behniden	1134	1137	1065	1137	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Erik_Emune	1100	18/09/1137
16	Erik 3. Lam	Behniden	1137	1146	1100	1146	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Erik_Lam	1120	27/08/1146
17	Svend 3.; Knud 5.; Valdemar 1	Behniden	1146	1157	1120	N/A	M	Jellingdynastiet	N/A	N/A	N/A
18	Valdemar 1. den Store	Behniden	1157	1182	1131	1182	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Valdemar_den_Store	14/01/1131	12/05/1182

Portfolio Element 4 - Text Mining in RStudio

Week 13 exercise. The task used the text of A Game of Thrones to practise PDF extraction, tokenisation, stop-word removal, word counting, and sentiment analysis. The submitted R Markdown file and RStudio project are included in the accompanying portfolio source folder.

```
library(tidyverse)
library(here)
library(pdftools)
library(tidytext)
library(textdata)

got_path <- here("got.pdf")
got_text <- pdf_text(got_path)

got_df <- data.frame(got_text) %>%
  mutate(text_full = str_split(got_text, pattern = "\n")) %>%
  unnest(text_full) %>%
  mutate(text_full = str_trim(text_full))

got_tokens <- got_df %>%
  unnest_tokens(word, text_full)

got_stop <- got_tokens %>%
  anti_join(stop_words)

got_swc <- got_stop %>%
  count(word) %>%
  arrange(-n)
```

The important step in this exercise is the change from a PDF into a structured list of words. Once the text is tokenised, common stop words can be removed and the remaining vocabulary can be counted. This is the same basic workflow later used in the final project, although the final project uses a much smaller and more controlled dataset.

Portfolio Element 4 - Output and Reflection

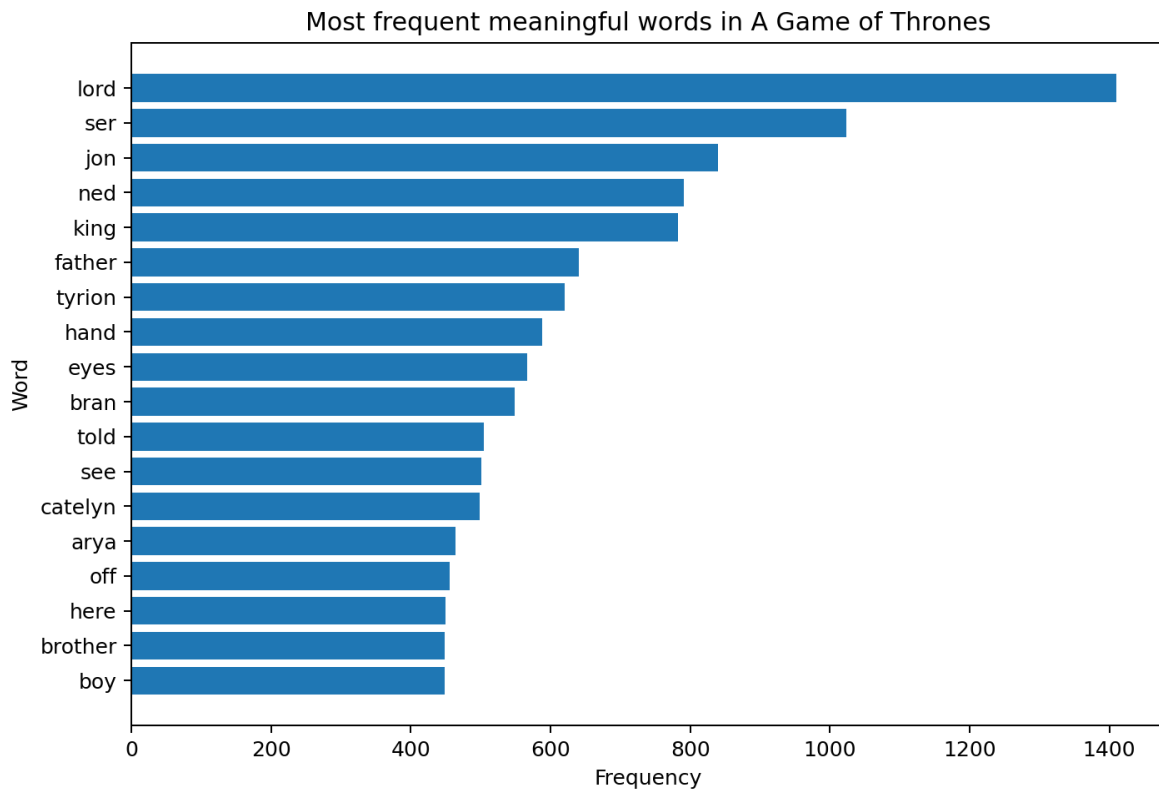


Figure P1. Frequent meaningful words in the submitted A Game of Thrones source after basic stop-word removal.

The output is dominated by character names, titles, and family relationships. This shows why word-frequency results have to be interpreted in relation to the source. A high count does not automatically make a word historically or thematically important; it may instead reflect the structure of a novel or the repeated appearance of major characters.

The exercise also introduced the AFINN sentiment lexicon. This helped prepare the later use of a lexicon-based sentiment method in the final project, while also showing a limitation: assigning a fixed score to individual words cannot fully recognise irony, context, or narrative voice.